WORKED SOLUTION

STEP III 2012, Question 6

Pure • Complex number

STEP III 2012, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/12-s3-q6

The key idea

A complex number is zero exactly when its real and imaginary parts are both zero, so one complex equation in $z = x + \mathrm{i}y$ becomes two real equations in x, y and p. The imaginary-part equation always factorises with a factor y, which splits every locus into "real roots" and "non-real roots". In each case we eliminate p to get a curve in the (x, y)-plane, and then check which points of that curve really come from a real value of p.

Solution

The first equation, $z^2 + pz + 1 = 0$

Put $z = x + \mathrm{i}y$ with x, y real. Then

$$z^2 + pz + 1 = (x^2 - y^2 + px + 1) + \mathrm{i}\,(2xy + py).$$

Since p is real, the two brackets are the real and imaginary parts, and both must vanish:

$$x^2 - y^2 + px + 1 = 0, \qquad (2x + p)y = 0,$$

as required.

From the second equation, either $p = -2x$ or $y = 0$. If $y = 0$, the first equation becomes $x^2 + px + 1 = 0$. Here $x \neq 0$, because $x = 0$ would give $1 = 0$, so we may divide by x to get $p = -(x^2 + 1)/x$. So either $p = -2x$, or $p = -(x^2 + 1)/x$ with $x \neq 0$ (and in this second case $y = 0$).

The root locus. *Case* $p = -2x$. Substituting into the real-part equation gives $x^2 - y^2 - 2x^2 + 1 = 0$, that is,

$$x^2 + y^2 = 1.$$

Conversely, take any point $x + \mathrm{i}y$ on the unit circle and set $p = -2x$, which is real. Then $(2x + p)y = 0$ and $x^2 - y^2 + px + 1 = 1 - x^2 - y^2 = 0$, so the point is a root. Hence the whole unit circle is in the locus.

Case $y = 0$, $p = -(x^2+1)/x$. For every real $x \neq 0$ this p is real, and x is a root of $x^2 + px + 1 = 0$. The point $x = 0$ is never a root, since $0^2 + p \cdot 0 + 1 = 1$.

So the root locus is the real axis with the origin missing, together with the unit circle $|z| = 1$. (The points ± 1 lie on both.)

The second equation, $pz^2 + z + 1 = 0$

With $z = x + iy$, the real and imaginary parts give

$$p(x^2 - y^2) + x + 1 = 0, \qquad (2px + 1)y = 0.$$

The origin is never a root, since $z = 0$ gives $1 = 0$.

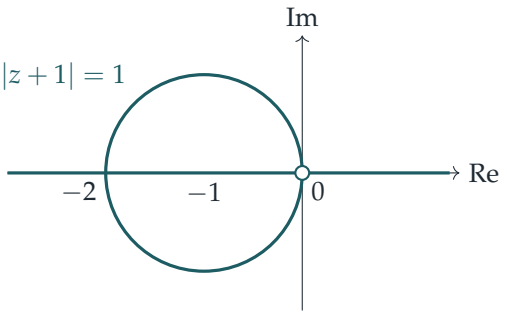
Case $y = 0$. Then $px^2 + x + 1 = 0$ with $x \neq 0$, so $p = -(x+1)/x^2$. This is real for every real $x \neq 0$, so every point of the real axis except the origin is in the locus. (When $x = -1$ we get $p = 0$, and the equation is the linear one $z + 1 = 0$, whose root is indeed -1.)

Case $2px + 1 = 0$. Then $x \neq 0$ and $p = -1/(2x)$. Substituting into the real-part equation and multiplying by $-2x$:

$$(x^2 - y^2) - 2x^2 - 2x = 0 \iff x^2 + y^2 + 2x = 0 \iff (x+1)^2 + y^2 = 1.$$

Conversely, for any point on this circle other than the origin we have $x \neq 0$, so $p = -1/(2x)$ is real, and reversing the steps shows both equations hold.

So the root locus is the real axis together with the circle $|z + 1| = 1$ (centre -1, radius 1), with the origin, which lies on both, removed.



The open circle marks the missing point $z = 0$.

Ask yourself Why does this locus look like the first one turned inside out?

If z is a root of $pz^2 + z + 1 = 0$ then $w = 1/z$ satisfies $w^2 + w + p = 0$. The map $z \mapsto 1/z$ sends the real axis (without 0) to itself and the line $\operatorname{Re} w = -\frac{1}{2}$, where the non-real roots of $w^2 + w + p = 0$ live, to the circle $|z + 1| = 1$ through the origin. The origin is missing because it would correspond to $w = \infty$.

The third equation, $pz^2 + p^2z + 2 = 0$

If $p = 0$ the equation reads $2 = 0$, which has no roots, so $p \neq 0$ throughout. The real and imaginary parts give

$$p(x^2 - y^2) + p^2x + 2 = 0, \qquad p(2x + p)y = 0,$$

and since $p \neq 0$ the second becomes $(2x + p)y = 0$.

Case $p = -2x$. Then $x \neq 0$, because $p \neq 0$. Substituting,

$$-2x(x^2 - y^2) + 4x^3 + 2 = 0 \quad \iff \quad x(x^2 + y^2) = -1.$$

Conversely, any point on this curve has $x \neq 0$, so $p = -2x$ is a non-zero real number, and reversing the steps shows it is a root. Since $x^2 + y^2 > 0$, the curve has $x < 0$. Writing it as

$$y^2 = -\frac{1}{x} - x^2,$$

we need $-1/x \geq x^2$; multiplying by $x < 0$ reverses the inequality to $-1 \leq x^3$, so $-1 \leq x < 0$. The curve passes through -1 (where $y = 0$), is symmetric in the real axis, and $y^2 \to \infty$ as $x \to 0^-$, so the imaginary axis is an asymptote. Also $\mathrm{d}(y^2)/\mathrm{d}x = x^{-2} - 2x > 0$ for $x < 0$, so $|y|$ increases steadily as x goes from -1 to 0. In polar form the curve is $r^3 \cos\theta = -1$.

Case $y = 0$. Now a real number x is in the locus when there is a real p with $xp^2 + x^2p + 2 = 0$. For $x = 0$ this says $2 = 0$, so 0 is not in the locus. For $x \neq 0$ it is a quadratic in p, with a real solution exactly when its discriminant is non-negative:

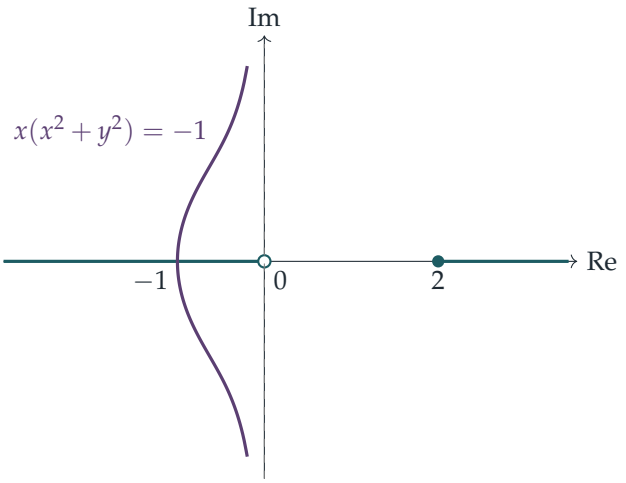
$$x^4 - 8x = x(x^3 - 8) \geq 0.$$

This holds when $x < 0$ (both factors negative) or when $x \geq 2$ (both factors non-negative), and fails for $0 < x < 2$. Any real p found this way is automatically non-zero, since $p = 0$ would give $2 = 0$. For example $x = 2$ needs $2p^2 + 4p + 2 = 0$, so $p = -1$, and indeed $-z^2 + z + 2 = -(z - 2)(z + 1)$.

Ask yourself Why solve for p here rather than for x?

We want to know which points x are attainable, so x is given and p is the unknown. The equation is quadratic in p, and the existence of a real root is exactly a discriminant condition on x.

So the root locus is the negative real axis, the ray $x \geq 2$ of the real axis, and the curve $x(x^2 + y^2) = -1$, which runs from -1 out towards the imaginary axis in both directions.



The curve meets the real axis only at -1, with a vertical tangent there, and approaches the imaginary axis as $|y| \to \infty$.

Checks

- For $z^2 + pz + 1 = 0$ the product of the roots is 1, so non-real roots are conjugates $z, \bar{z}$ with $|z|^2 = 1$: the unit circle again.
- In the third equation, $p = -1$ gives roots 2 and -1, the boundary point of the ray and the vertex of the curve. Taking $p = 1$ gives $z^2 + z + 2 = 0$, with roots $-\frac{1}{2} \pm \frac{\sqrt{7}}{2}\mathrm{i}$, and $x(x^2 + y^2) = -\frac{1}{2} \cdot 2 = -1$, as it should.
- Numerically, we solved all three equations for 10^5 random values of p and confirmed every root lies on the stated locus. We also confirmed that random points of each locus are attained with the stated p, and a fine scan of p found no real root of the third equation in $(0, 2)$; the smallest positive one was 2.0.

Common slips

- Forgetting to check the converse: eliminating p gives a curve that *contains* the locus, and you still have to show each point is reached by a real p. In the third equation, real points in $(0, 2)$ satisfy $y = 0$ but are never reached.
- Dividing by x or p without saying why it is non-zero. This is exactly where the missing origin comes from.
- Treating $p = 0$ carelessly: in the second equation it gives the genuine root -1, but in the third it gives no roots at all.

Beyond STEP

Root loci are a standard tool in control engineering: as a gain parameter varies, the poles of a system move along curves, and stability depends on whether they cross into the right half-plane. The question also hints at the geometry of the map $z \mapsto 1/z$, which sends lines and circles to lines and circles. This is the start of the theory of Möbius transformations, beautifully illustrated in Needham, *Visual Complex Analysis*.

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